

The Informational Resolution: Solving Modern Cosmological Paradigms via the HIGM Framework

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(This model was developed in the framework of theoretical physics and AI)

Abstract:

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Modern cosmology faces a "Crisis of Concordance," characterized by the Hubble Tension, the Cosmological Constant Problem, and the unexpected maturity of early galaxies. This paper demonstrates that the Hamouda Informational Gravity Model (HIGM) resolves these anomalies by redefining the vacuum as an active informational substrate. By applying Shannon Entropy to gravitational dynamics, we provide a unified solution to the Baryon Asymmetry (Matter-Antimatter) problem, the Lithium Problem, and the "Fine-Tuning" paradox, offering a testable alternative to Dark Matter and Dark Energy.

Keywords: *HIGM, Informational Vacuum, Hubble Tension, Baryon Asymmetry, Cosmological Constant Problem, Shannon Entropy, Radiative Genesis, Dark Sector Alternative.*

1. Introduction

Traditional cosmology relies on "Dark" components—Dark Matter and Dark Energy—which remain undetected despite decades of searching. The HIGM proposes that these are not physical substances but emergent behaviors of a dense Informational Vacuum. In this framework, gravity is a thermodynamic "overhead" required processing the Infrared Information Flux (Φ_{IR}) of the universe. This transition from a "Hardware-based" (Matter)

to a "Software-based" (Information) universe allows for the resolution of deep-seated paradoxes that standard General Relativity (GR) cannot address.

2. Mathematical Framework: The Entropic Bridge

The core of the HIGM is the relationship between vacuum information density (ρ_{info}) and gravitational curvature. The model derives the 9.05% Informational Boost β from the Shannon Entropy formula:

$$g_{\text{HIGM}} = g_{\text{GR}} (1 + \beta)$$

Where β is a dynamic variable governed by the local information flux. This fundamental coupling allows the HIGM to solve problems across all cosmic scales.

3. HIGM Contributions to Unsolved Problems

3.1. The Hubble Tension (Expansion Discrepancy)

- The Problem: Local measurements ($H_0 \sim 73$) disagree with early-universe CMB data ($H_0 \sim 67$).
- The HIGM Solution: HIGM treats expansion as a thermodynamic response. The "tension" is actually the evolution of entropy. In the low-entropy early universe, the vacuum processed data slower ($H_0=67$). As stars formed and emitted IR flux, the vacuum's information density increased, "accelerating" the expansion to the current 73.5 km/s/Mpc.

3.2. Baryon Asymmetry (Matter vs. Antimatter)

- The Problem: Why did matter survive while antimatter disappeared?
- The HIGM Solution (Bit-Asymmetry): During the Radiative Genesis, the vacuum found the matter-encoding format slightly more "informationally efficient" (10^{-9} bit). Matter became the "Default Save Format" of the vacuum's memory. Annihilation is a "Bit-Reset" returning data to the massless "Data Bus" (Light).

3.3. The Cosmological Constant Problem (10^{120} discrepancy)

- The Problem: Quantum predictions of vacuum energy are 120 orders of magnitude too high.
- The HIGM Solution: "Vacuum Energy" is not a physical weight but Processing Capacity. HIGM replaces the energy density with Shannon Information Density. The observed value is simply the current "Saturation Limit" of the universe's data processing, naturally matching the low observed value without fine-tuning.

3.4. The Lithium Problem (BBN Paradox)

- The Problem: The Big Bang produced 3x more Lithium-7 than we see in old stars.
- The HIGM Solution: The 9.05% Informational Boost altered the expansion rate during the first minutes (Big Bang Nucleosynthesis). This "Entropic Acceleration" shifted the cooling curve of the Pre-Bang Light, naturally suppressing Lithium-7 production to match observed levels.

3.5. The "Impossible" Early Galaxies (JWST)

- The Problem: JWST sees mature, massive galaxies that shouldn't exist so early.
- The HIGM Solution: In the low-entropy early universe, the Informational Vacuum acted as a Gravitational Superconductor. Matter collapsed into structure 3.5x faster than standard models allow, because the vacuum didn't need "Dark Matter" to pull gas together; the entropic force was already at maximum efficiency.

4. Conclusion

The HIGM demonstrates that the "unsolved" problems of cosmology are symptoms of a missing paradigm. By recognizing light as the "Data Bus" and gravity as "Entropy Friction," we align the laws of the universe with the laws of Information Theory. The model provides a stable, predictive, and mathematically elegant roadmap for 21st-century physics.

5. References

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